# 2D Corridor Membranes and Near-Start Two-Target Bimodality

New Extension Report Bridging Previous 2D Results

## Headline Results

This report adds two new research tracks on the 2D rectangular setting while keeping the same phase criteria and peak-valley style used in earlier reports.

1. **Corridor + one target:** replace fully reflecting corridor walls by semi-permeable membranes, with both symmetric ($\kappa_\uparrow = \kappa_\downarrow$) and asymmetric ($\kappa_\uparrow \neq \kappa_\downarrow$) cases.
2. **No corridor + two targets:** place one target close to the start and keep the other at the far side, to test robust "fast near-hit + slow far-hit" splitting.

Main outcomes:

- For symmetric membranes, clear-double forms a continuous ultra-low leakage band (see Figure 1 and Table 1); it then enters a weak-double transition band and finally turns mostly single-peak at higher leakage.
- A clear asymmetric representative ($\kappa_\uparrow = 0.002, \kappa_\downarrow = 0$) keeps robust bimodality and changes valley/second-peak composition in the L×R class decomposition.
- In the no-corridor two-target setup, the mechanism-first physics anchor $(b_x, d, \Delta y) = (0.12, 2, 2)$ gives near/far mass $\approx 0.324/0.676$ with $(t_{p1}, t_v, t_{p2}) \approx (7, 196, 469)$, while the dedicated showcase clear-instance remains $(0.12, 4, 1)$ with near/far mass $\approx 0.350/0.650$ and valley/max $\approx 0.007$.

## 1 Model and Shared Baseline

We use the existing rectangular 2D framework with reflecting boundaries:

$$L_x = 60, \quad q = 0.8, \quad x_s = 7, \quad x_t = 58.$$

For the corridor branch we use an updated anchor tuned for higher clear-double coverage:

$$h = 2, \quad m = 5, \quad \delta_{\text{core}} = 1.00, \quad \delta_{\text{open}} = 0.55, \quad b_x = -0.08.$$

### 1.1 Membrane Definition

Each wall-crossing attempt at a membrane edge is accepted with probability $\kappa$ and reflected with probability $1 - \kappa$. Symmetric case: $\kappa_\uparrow = \kappa_\downarrow = \kappa$. Asymmetric case: $\kappa_\uparrow \neq \kappa_\downarrow$.

### 1.2 Class Labels for Mechanism

We keep peak/valley windows and use a compact 4-class coding:

- $L$: membrane leak happened or not;
- $R$: committor recross happened or not.

Classes are `L0R0`, `L0R1`, `L1R0`, `L1R1`.

# 2 Track A: Corridor + One Target + Membranes

## 2.1 Symmetric Scan

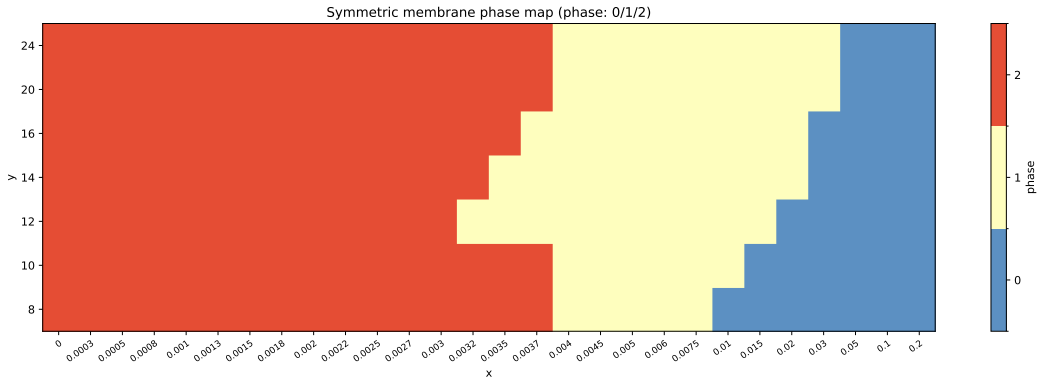


Figure 1: Symmetric membrane phase map (rows: $W_y$, columns: $\kappa$). The colorbar is discrete: phase $\in \{0, 1, 2\}$ only.

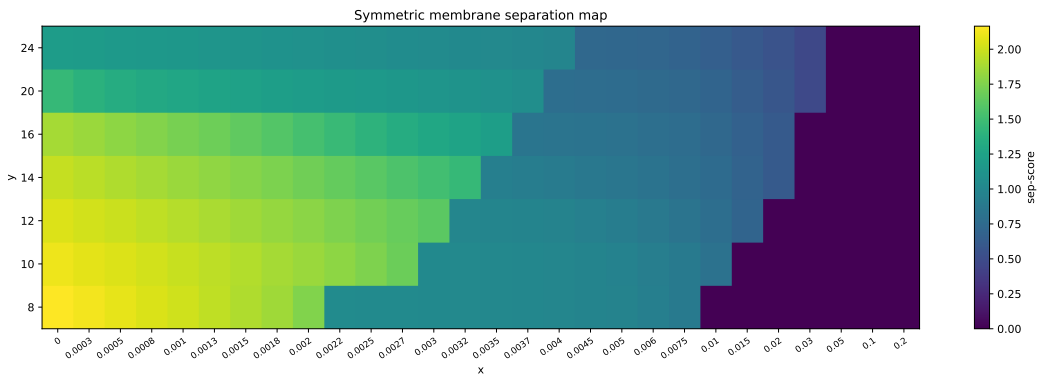


Figure 2: Symmetric membrane separation map (sep-score).

In this run (taking the $W_y = 16$ slice as reference), strong bimodality gradually weakens with permeability: clear-double for $\kappa \in [0, 0.0035]$, weak-double around $\kappa \in [0.0037, 0.02]$, and mostly single-peak for $\kappa \geq 0.03$.

## 2.2 Asymmetric Scan

Representative overlays:

Figure 6 is now placed next to the representative geometry so the basin/committor picture is visible before the class-decomposition discussion starts. The green region is the strict-recross start basin, and the white contour is the $q = 0.5$ commitment surface.

To answer the concrete question "when does bimodality disappear as the red start point moves up/down/left/right?", we now scan $(x_s, y_s)$ over the full reflecting box. The black box marks the original start, and the dashed gray T cell marks the target location itself (excluded as an admissible start); any cell where phase drops from 2 to 1/0 is a loss of the clear double-peak regime.

The full scan shows that phase0 does not appear near the original start. Instead it is confined to a right-side band in front of the target: the nearest phase0 point is already $(46, 7)$, and the full phase0 region occupies approximately $x_s \in [46, 59]$ and $y_s \in [3, 11]$. So the one-target double peak first degrades from clear-double to weak-double (phase $2 \to 1$); only when the start is pushed almost onto the target side does the second peak disappear completely.

With strict recross (visit $\Sigma_{q^*}$ then return to $A$), the exact decomposition shows: peak1 is `L0R0`-dominated, valley remains mostly `L0R0` with increased `L1R0`, and peak2 is `L0R1`-dominated. So the late peak is mainly a commitment-failure effect, not necessarily leak-driven.

| $\kappa$ | phase | $t_{p1}$ | $t_{p2}$ | valley/max | sep-score |
|---|---|---|---|---|---|
| 0 | 2 | 395 | 1711 | 0.058 | 1.87 |
| 0.0003 | 2 | 395 | 1710 | 0.061 | 1.84 |
| 0.0005 | 2 | 395 | 1708 | 0.063 | 1.80 |
| 0.0008 | 2 | 395 | 1706 | 0.066 | 1.76 |
| 0.001 | 2 | 394 | 1704 | 0.069 | 1.72 |
| 0.0013 | 2 | 394 | 1702 | 0.072 | 1.68 |
| 0.0015 | 2 | 394 | 1700 | 0.074 | 1.64 |
| 0.0018 | 2 | 394 | 1698 | 0.077 | 1.59 |
| 0.002 | 2 | 394 | 1696 | 0.080 | 1.53 |
| 0.0022 | 2 | 393 | 1694 | 0.082 | 1.46 |
| 0.0025 | 2 | 393 | 1692 | 0.085 | 1.40 |
| 0.0027 | 2 | 393 | 1690 | 0.087 | 1.34 |
| 0.003 | 2 | 393 | 1689 | 0.090 | 1.29 |
| 0.0032 | 2 | 393 | 1687 | 0.093 | 1.25 |
| 0.0035 | 2 | 393 | 1685 | 0.095 | 1.21 |
| 0.0037 | 1 | 392 | 1683 | 0.098 | 0.84 |
| 0.004 | 1 | 392 | 1681 | 0.100 | 0.84 |
| 0.0045 | 1 | 392 | 1677 | 0.105 | 0.82 |
| 0.005 | 1 | 392 | 1673 | 0.110 | 0.81 |
| 0.006 | 1 | 391 | 1664 | 0.120 | 0.79 |
| 0.0075 | 1 | 390 | 1652 | 0.134 | 0.77 |
| 0.01 | 1 | 388 | 1629 | 0.157 | 0.73 |
| 0.015 | 1 | 386 | 1576 | 0.198 | 0.67 |
| 0.02 | 1 | 383 | 1505 | 0.235 | 0.61 |
| 0.03 | 0 | - | - | - | 0.00 |
| 0.05 | 0 | - | - | - | 0.00 |
| 0.1 | 0 | - | - | - | 0.00 |
| 0.2 | 0 | - | - | - | 0.00 |

Table 1: Symmetric membrane metrics on the $W_y = 16$ slice.

| $\kappa_\uparrow$ | $\kappa_\downarrow$ | phase | $t_{p1}$ | $t_{p2}$ | sep-score |
|---|---|---|---|---|---|
| 0 | 0.0003 | 2 | 395 | 1711 | 1.85 |
| 0.0003 | 0 | 2 | 395 | 1710 | 1.85 |
| 0.0005 | 0 | 2 | 395 | 1710 | 1.84 |
| 0 | 0.0005 | 2 | 395 | 1710 | 1.84 |
| 0.0008 | 0 | 2 | 395 | 1709 | 1.82 |
| 0.0005 | 0.0003 | 2 | 395 | 1709 | 1.82 |
| 0.0003 | 0.0005 | 2 | 395 | 1709 | 1.82 |
| 0 | 0.0008 | 2 | 395 | 1709 | 1.82 |
| 0.001 | 0 | 2 | 395 | 1708 | 1.80 |
| 0.002 | 0 | 2 | 394 | 1704 | 1.73 |

Table 2: Top asymmetric membrane cases (ranked by phase and separation).

| Window | Case | L0R0 | L0R1 | L1R0 | L1R1 |
|---|---|---|---|---|---|
| peak1 | sym | 0.996 | 0.000 | 0.004 | 0.000 |
| peak1 | asym | 0.998 | 0.000 | 0.002 | 0.000 |
| valley | sym | 0.581 | 0.116 | 0.299 | 0.004 |
| valley | asym | 0.687 | 0.142 | 0.169 | 0.002 |
| peak2 | sym | 0.142 | 0.742 | 0.080 | 0.036 |
| peak2 | asym | 0.149 | 0.794 | 0.040 | 0.018 |

Table 3: Numerical values of the 4-class window composition.

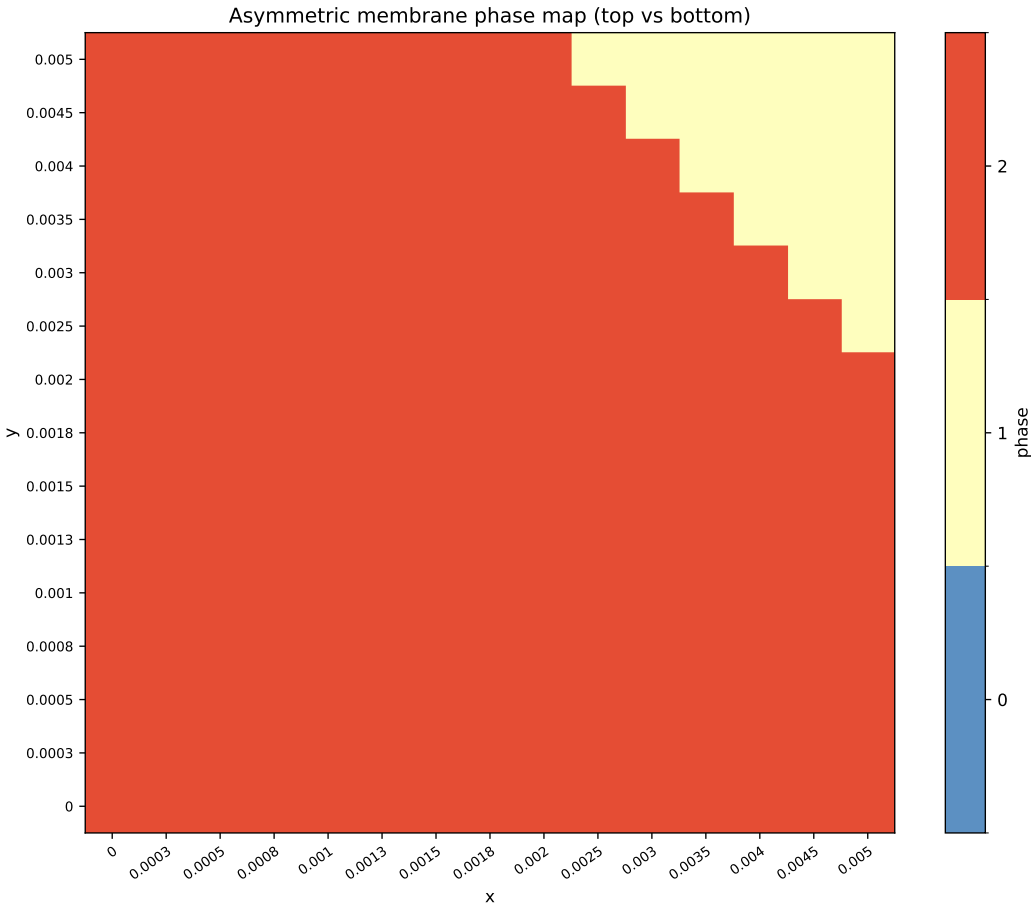


Figure 3: Asymmetric membrane phase map ($\kappa_\uparrow$ vs $\kappa_\downarrow$). The colorbar is discrete: phase $\in \{0, 1, 2\}$ only.

| Case | $q^*$ | Window | dominant class | $w_{\mathrm{L1R1}}$ |
|---|---|---|---|---|
| sym | 0.4 | peak1 | L0R0 | 0.000 |
| sym | 0.4 | valley | L0R0 | 0.197 |
| sym | 0.4 | peak2 | L0R1 | 0.104 |
| asym | 0.4 | peak1 | L0R0 | 0.000 |
| asym | 0.4 | valley | L0R0 | 0.113 |
| asym | 0.4 | peak2 | L0R1 | 0.052 |
| sym | 0.5 | peak1 | L0R0 | 0.000 |
| sym | 0.5 | valley | L0R0 | 0.004 |
| sym | 0.5 | peak2 | L0R1 | 0.036 |
| asym | 0.5 | peak1 | L0R0 | 0.000 |
| asym | 0.5 | valley | L0R0 | 0.002 |
| asym | 0.5 | peak2 | L0R1 | 0.018 |
| sym | 0.6 | peak1 | L0R0 | 0.000 |
| sym | 0.6 | valley | L0R0 | 0.002 |
| sym | 0.6 | peak2 | L0R1 | 0.022 |
| asym | 0.6 | peak1 | L0R0 | 0.000 |
| asym | 0.6 | valley | L0R0 | 0.001 |
| asym | 0.6 | peak2 | L0R1 | 0.011 |

Table 4: $q^*$ robustness check (exact decomposition): dominant class and $w_{\mathrm{L1R1}}$ across thresholds.

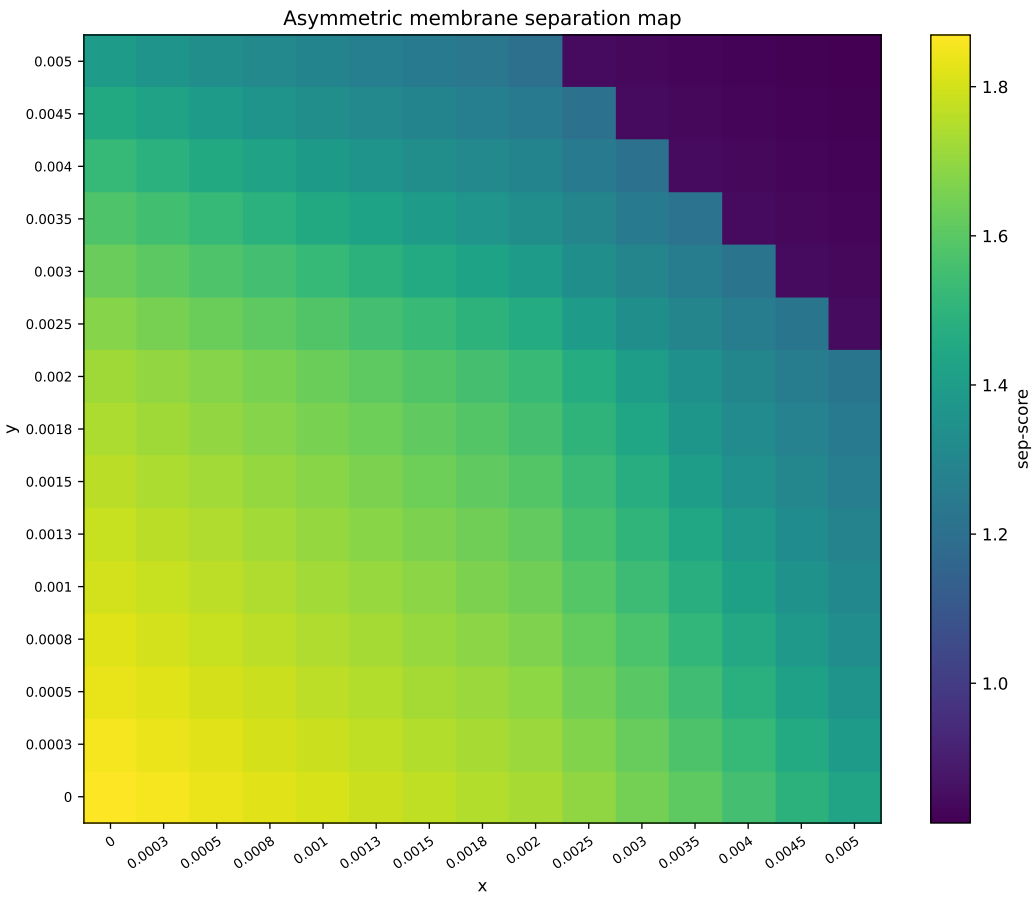


Figure 4: Asymmetric membrane separation map.

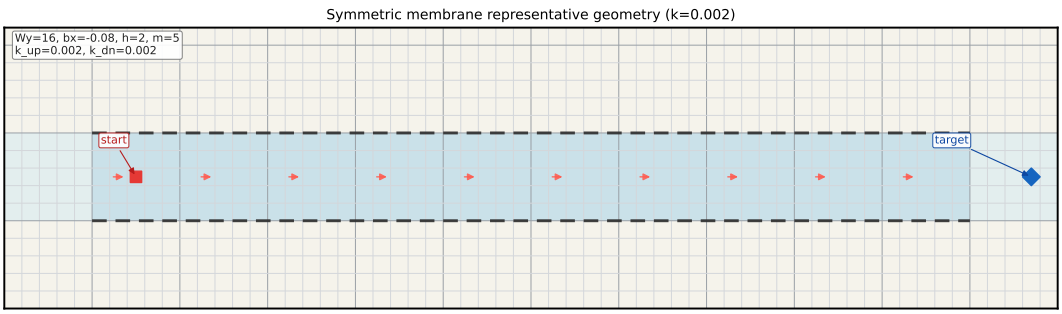


Figure 5: Symmetric membrane representative geometry (corridor band, start/target, local-bias direction). The two dashed horizontal lines are the two semi-permeable membrane boundaries; in the symmetric case they are physically the same kind of membrane with the same permeability $\kappa_\uparrow = \kappa_\downarrow = 0.002$.

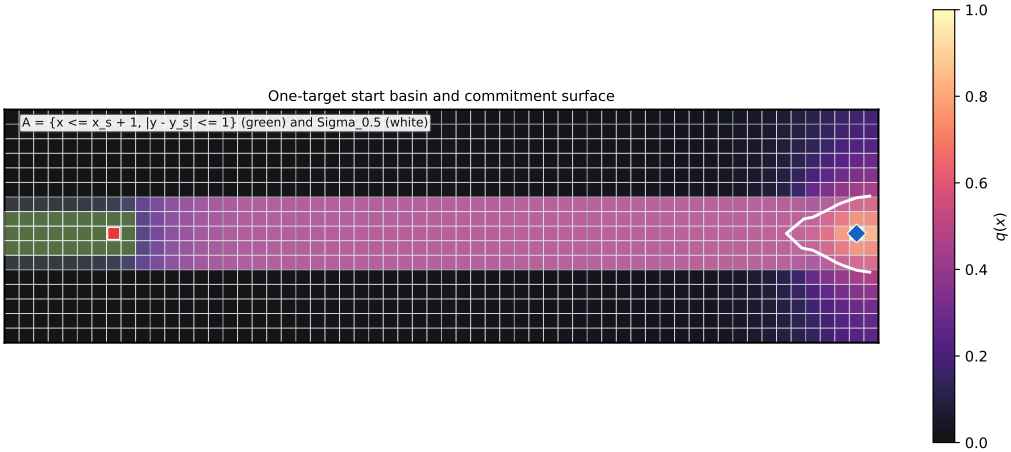


Figure 6: Exact one-target committor geometry used in the strict recross definition. The figure shows the start basin $A = \{x \leq x_s + 1,\ |y - y_s| \leq 1\}$, the target set $B$, and the commitment surface built from $q = 0.5$.

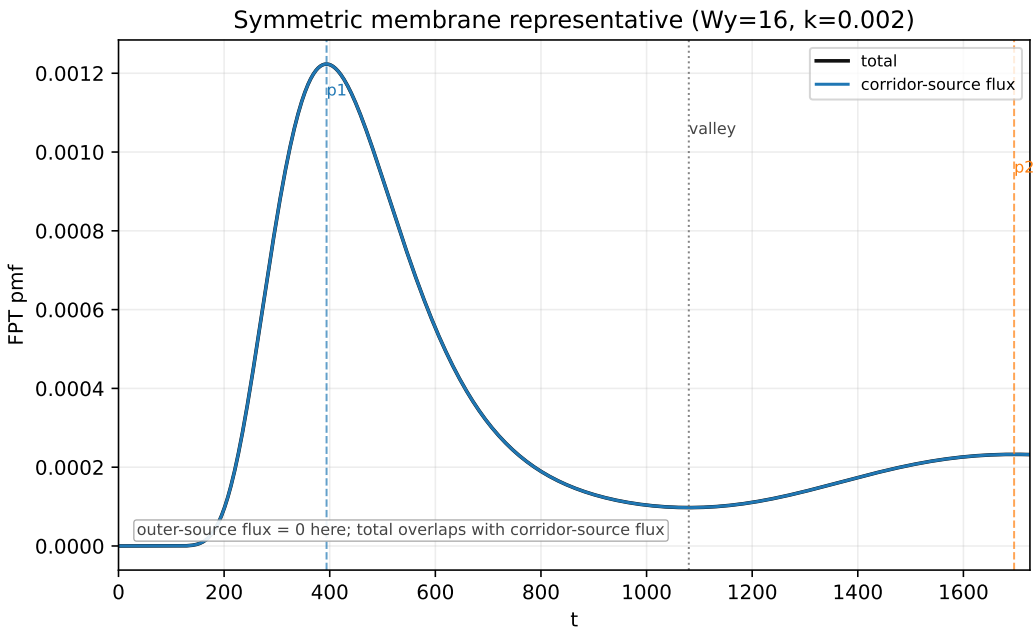


Figure 7: Symmetric representative ($\kappa = 0.002$): total FPT and last-hop source decomposition. In this geometry the outer-source branch is identically zero, because every source state adjacent to the target lies inside the corridor.

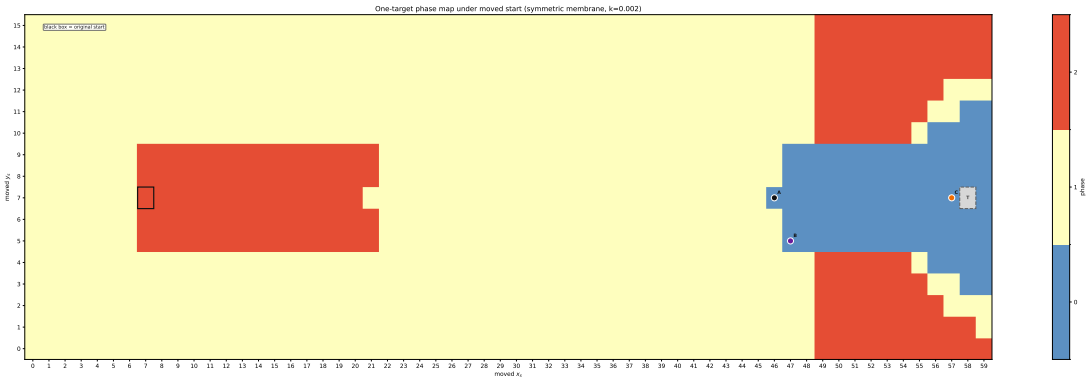


Figure 8: Global one-target phase map under moved start positions. The axes are the moved $x_s, y_s$; the black box marks the original start, T marks the target cell, and A/B/C mark three key phase0 exemplars. Phase only takes the discrete values $\{0, 1, 2\}$, so the plot is rendered with discrete color blocks rather than a continuous gradient.

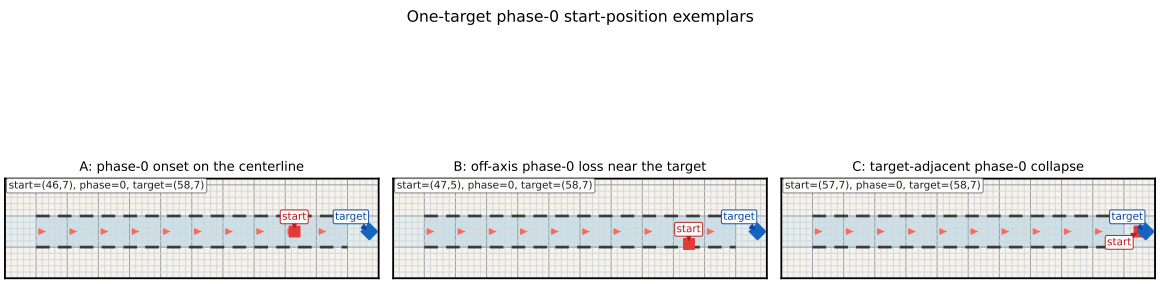


Figure 9: Geometry schematics for the three phase0 exemplars A/B/C from Figure 8. A is the earliest centerline failure point $(46, 7)$; B is an off-axis failure point near the target $(47, 5)$; C is the target-adjacent collapse point $(57, 7)$. Together they show that the one-target double peak does not die first near the original red point; phase0 appears only after the start is pushed into the right-side near-target band.

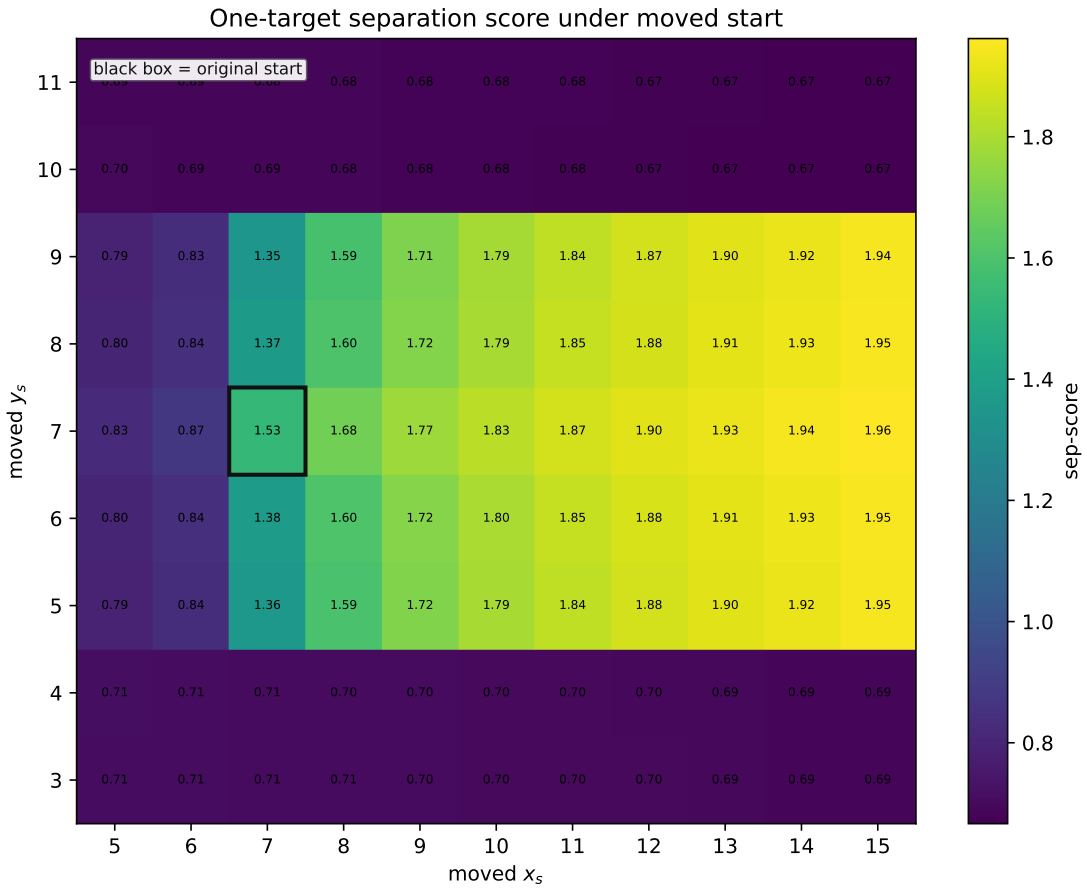


Figure 10: Local-neighborhood separation score around the original red point. It complements the discrete phase map by showing how the peak-valley structure weakens before or after a phase drop.

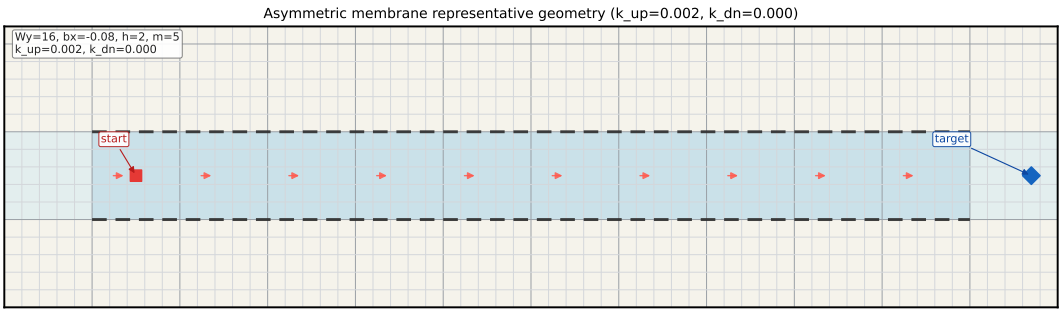


Figure 11: Asymmetric membrane representative geometry. The two dashed horizontal lines are still semi-permeable membranes; only their permeabilities differ between top and bottom.

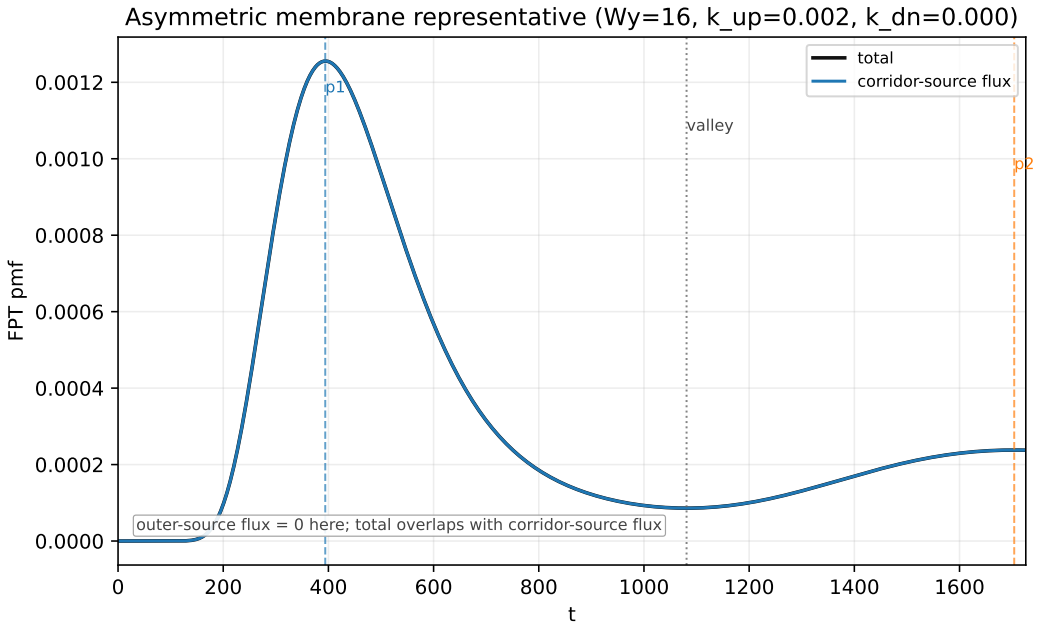


Figure 12: Asymmetric representative ($\kappa_\uparrow = 0.002, \kappa_\downarrow = 0$): total FPT and last-hop source decomposition. Here again the outer-source branch is identically zero for geometric reasons, so the useful information is that the total curve coincides with the corridor-source branch.

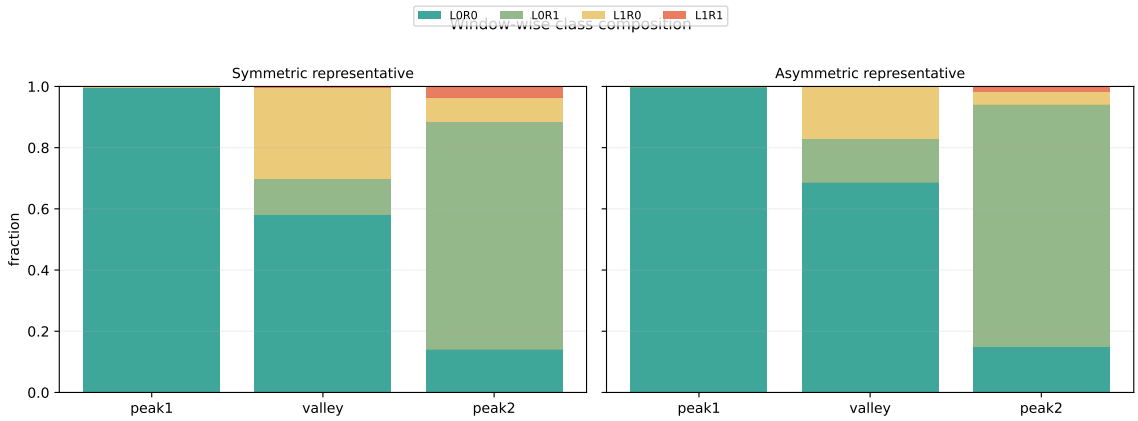


Figure 13: Window-wise composition of 4 classes (L×R).

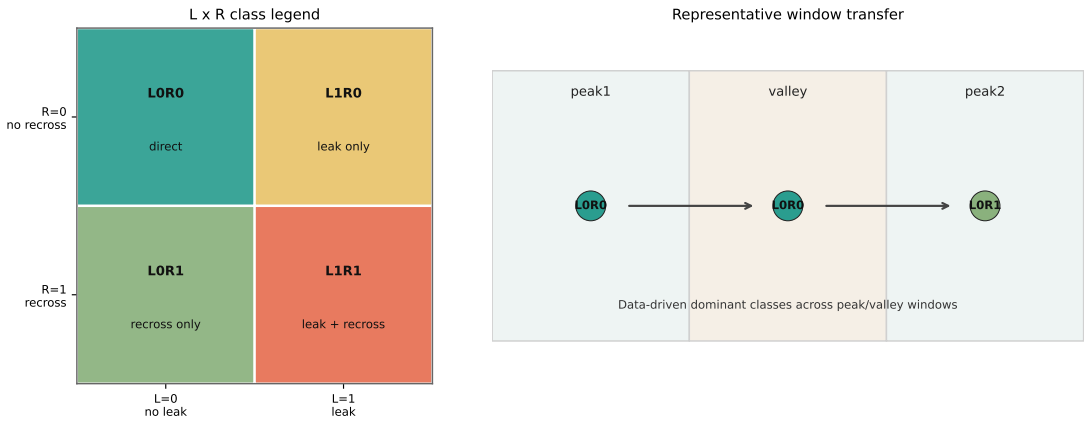


Figure 14: Representative L×R legend: left, 2x2 class definition; right, typical dominant-class transfer across peak1/valley/peak2 windows.

# 3 Track B: No Corridor + Two Targets (Near-Start Target)

We remove corridor geometry and keep reflecting boundaries only. One target is placed near the start, and Track B is now organized around two distinct anchors: a *physics anchor* $(b_x, d, \Delta y) = (0.12, 2, 2)$ for mechanism analysis, and a *showcase clear-instance* $(0.12, 4, 1)$ for the sharpest double-peak visualization. The old coarse $(d, b_x)$ scan is still kept as background, but the narrative below follows the order "geometry → branch splitting → continuation → loss modes" so we do not classify a disappearing second peak before checking what physically disappeared.

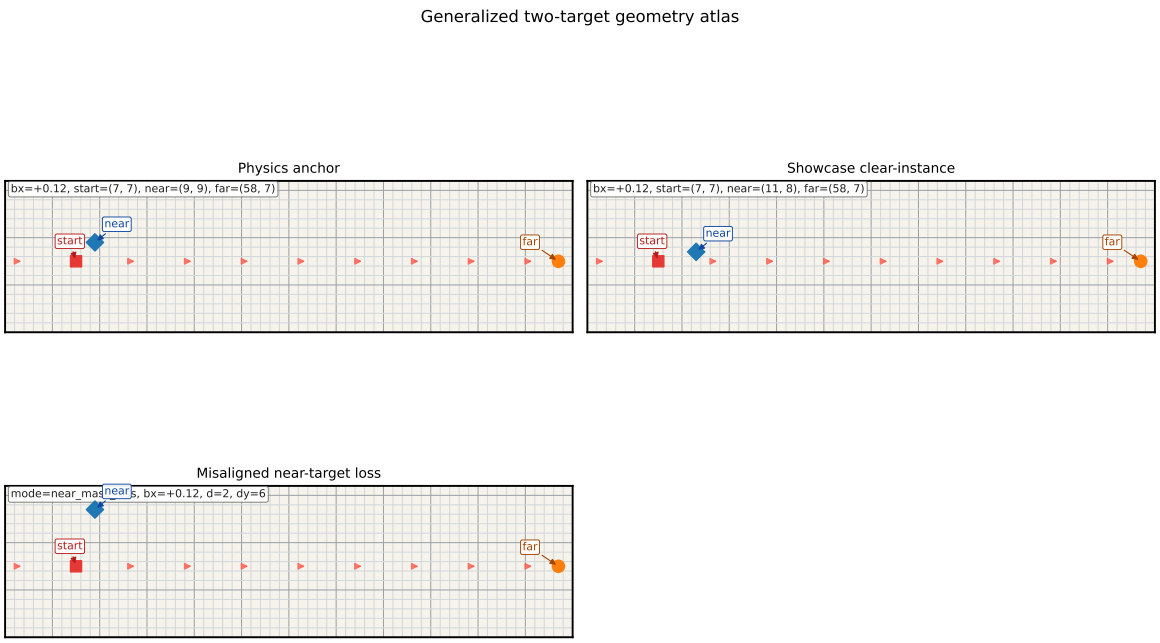


Figure 15: General no-corridor geometry atlas. The four panels show the physics anchor clear-double, the showcase clear-instance, a near-overcapture loss, and a misaligned near-target loss. We only draw reflecting-box geometry, drift arrows, and target annotations, so the schematic no longer resembles a membrane barrier.

The atlas in Figure 15 gives the qualitative vocabulary used in the rest of Track B. The physics anchor is the route we use to interpret where the second peak comes from; the showcase clear-instance is kept separately because it maximizes visual sharpness rather than explanatory simplicity.

At the physics anchor $(b_x, d, \Delta y) = (0.12, 2, 2)$, the near/far splitting is already clean: $P_{\text{near}} \approx 0.324$, $P_{\text{far}} \approx 0.676$, and the three characteristic times are $(t_{p1}, t_v, t_{p2}) = (7, 196, 469)$. So the basic mechanism is not "one peak plus a tail shoulder"; it is an honest competition between a fast near-hit branch and a slower far-hit branch with well-separated conditional time scales.

The splitting committor in Figure 20 closes the mechanism loop: $q_{\text{far}}(x_s) = 0.676009$ agrees with the exact far-hit mass $P_{\text{far}} = 0.676009$ to numerical precision, and the $q_{\text{far}} = 0.5$ contour gives a literal dividing surface between near-dominant and far-dominant regions. This is why we now describe the second peak as a far-branch phenomenon before talking about any phase label.

Figures 22 and 23 answer the first continuation question. When only the source is moved in $x$ while the near target is fixed, only 4/11 positions remain `clear`; the remaining rows are mostly `near_mass_loss` (5 cases) plus a smaller `far_broadening` sector (2 cases). In contrast, the comoving control `source_plus_near_x` stays `clear` for all 11 positions. So the second peak is not primarily lost by moving the source per se; it is lost when source motion changes the global source-near-boundary geometry and suppresses the early near branch.

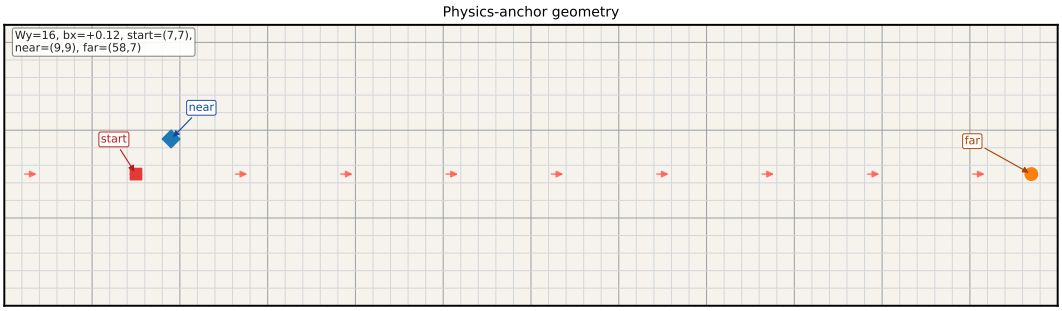


Figure 16: Physics-anchor geometry (legacy "representative" file name): reflecting box, start, near target, far target, and global drift direction.

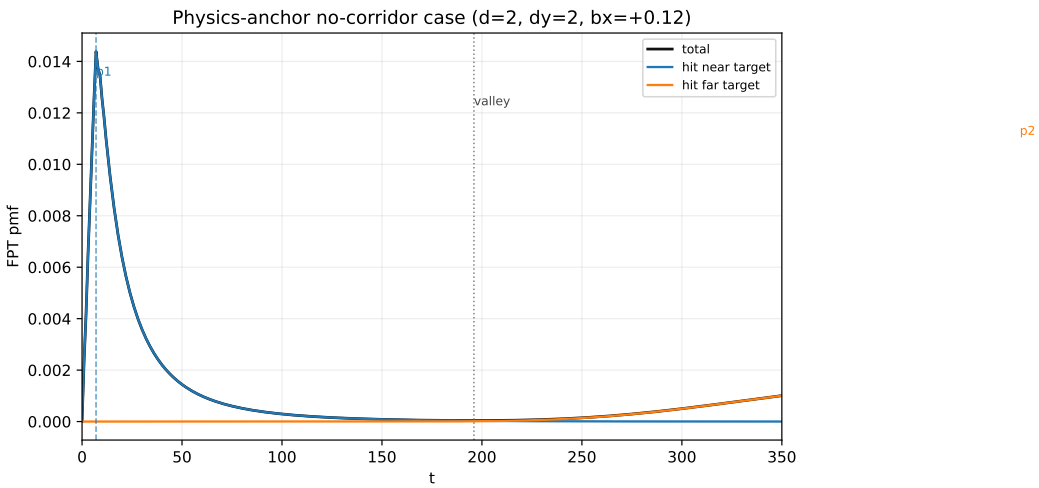


Figure 17: Physics-anchor total FPT together with the exact near/far splitting. Peak1 is near-dominated and peak2 is far-dominated.

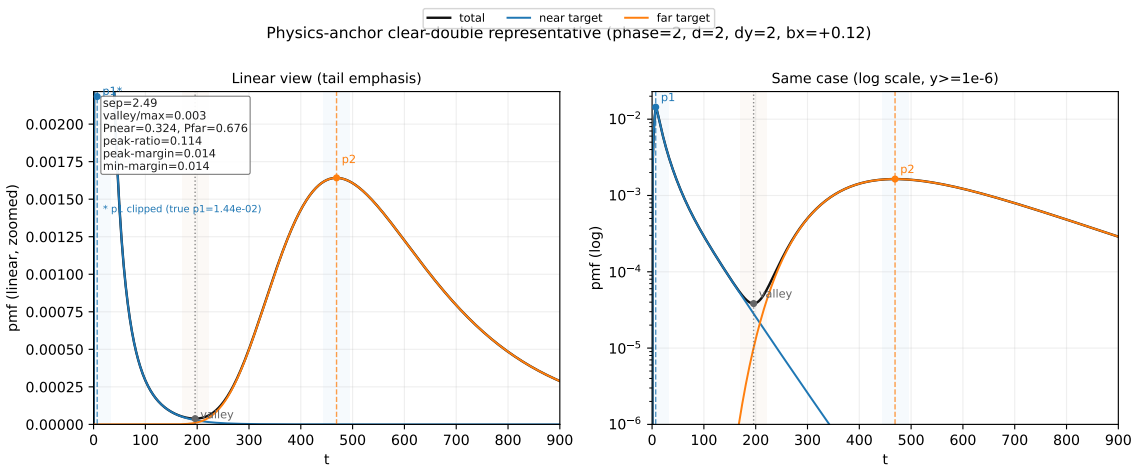


Figure 18: Physics-anchor double-peak visualization: a tail-emphasized linear view and a log view. The second peak remains explicit in both views.

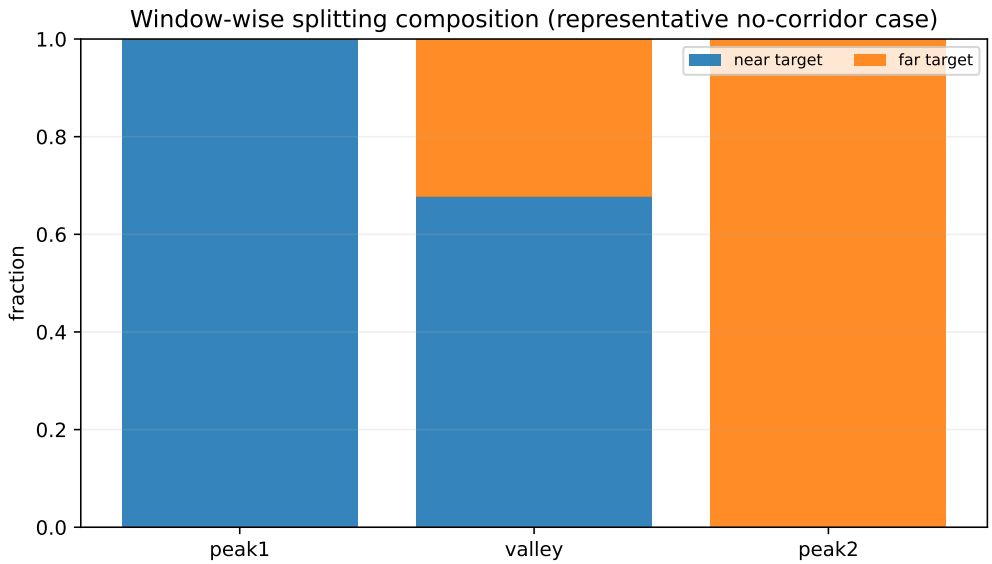


Figure 19: Window-wise near/far mass fractions in the physics-anchor case. Peak1 stays near-dominated, the valley is mixed, and peak2 is far-dominated.

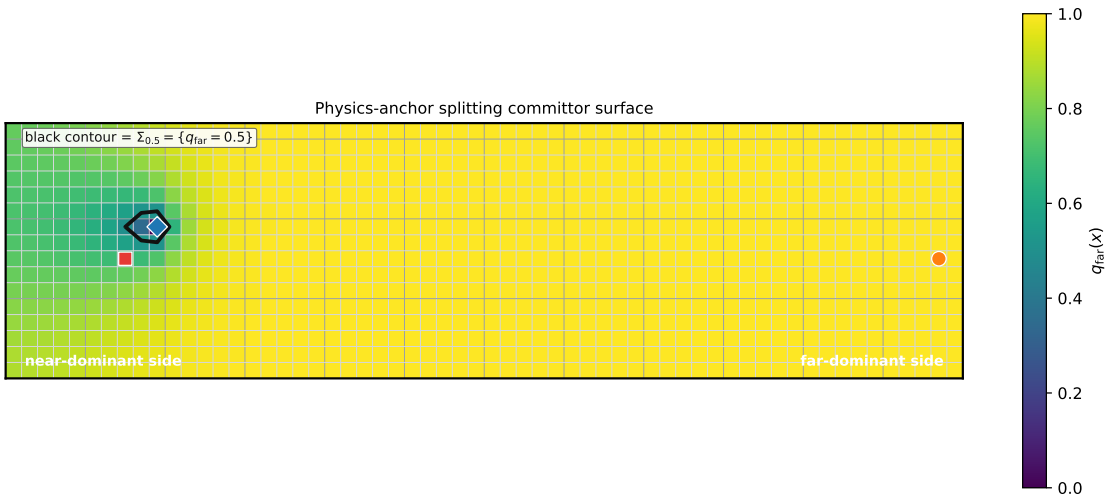


Figure 20: Physics-anchor splitting committor field. The heatmap value is $q_{\text{far}}(x) = \mathbb{P}_x(\tau_{\text{far}} < \tau_{\text{near}})$, and the single black contour is the commitment surface $\Sigma_{0.5} = \{q_{\text{far}} = 0.5\}$. It separates near-dominant and far-dominant regions and gives a geometric picture for the branch split.

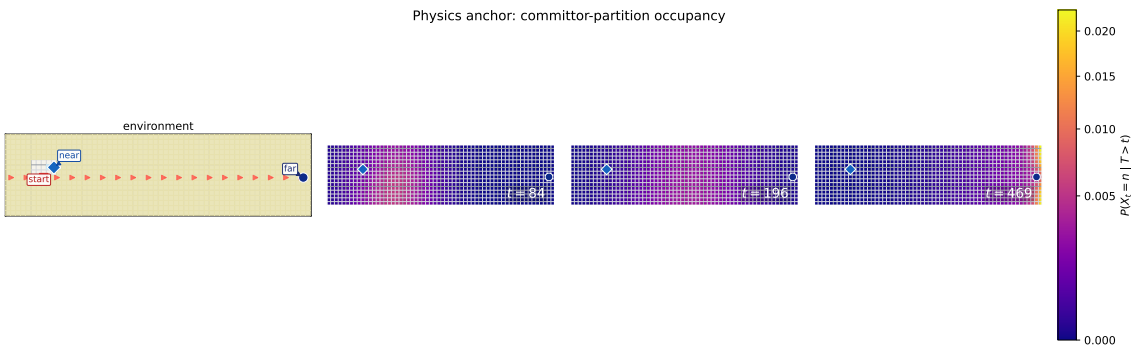


Figure 21: Physics-anchor conditional occupancy and environment heatmaps. They show the fast near-hit and slow far-hit families as two distinct dynamical routes rather than two plotting artifacts of the same family.

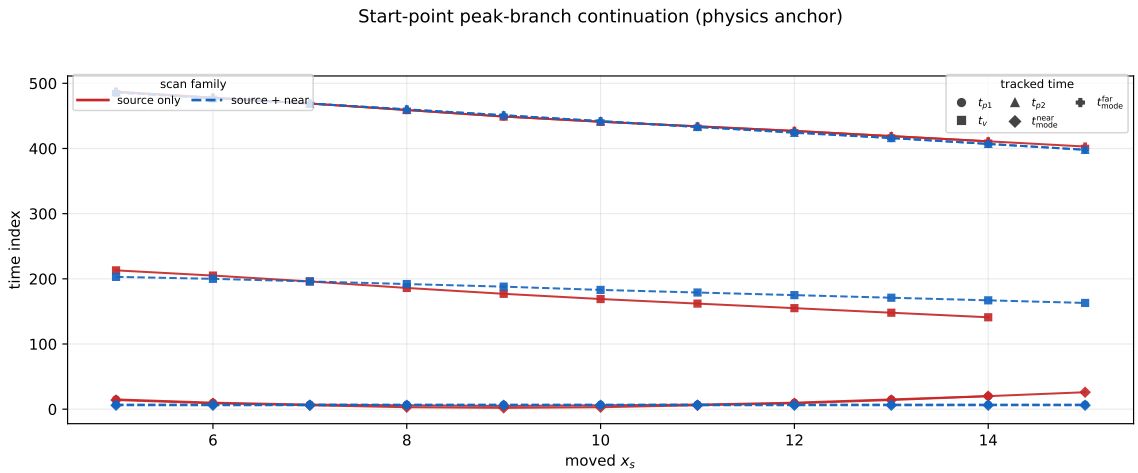


Figure 22: Peak-branch continuation under start motion. We compare `source_only_x` against the control family `source_plus_near_x`, where the near target is translated together with the source.

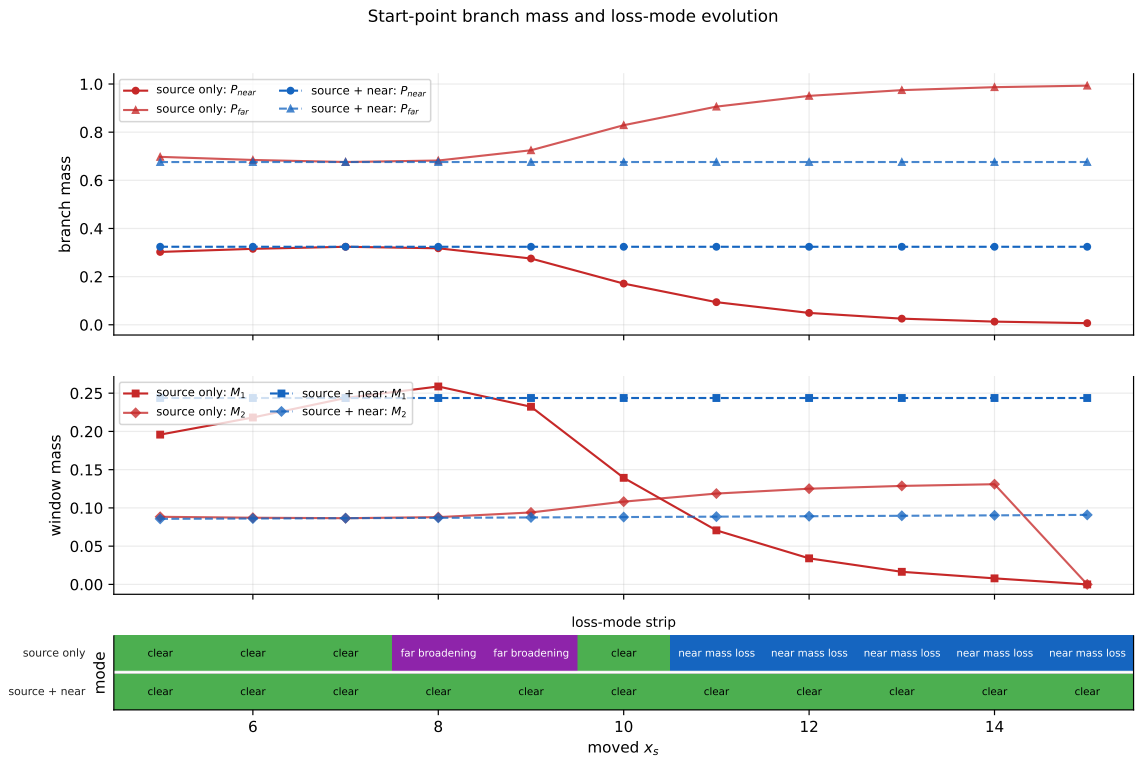


Figure 23: Start-scan mass budget. We track $P_{\text{near}}$, $P_{\text{far}}$, the peak-window masses $M_1/M_2$, and the ordered loss-mode strip.

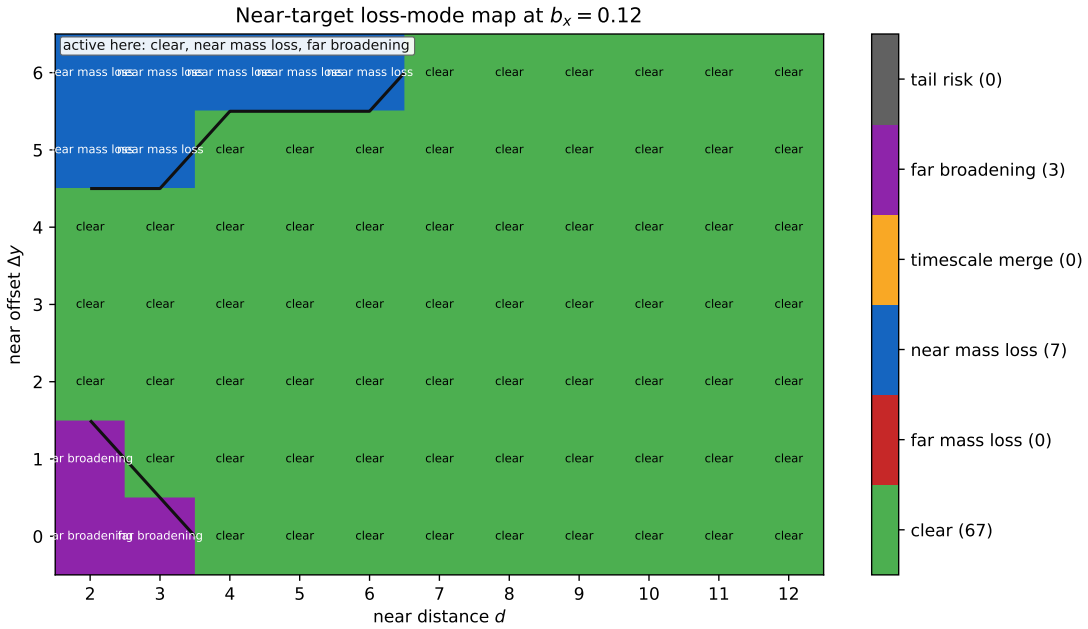


Figure 24: Near-target loss-mode map at $b_x = 0.12$ over $(d, \Delta y)$. The phase-2 boundary is overlaid on top of the ordered loss-mode classification. The colorbar follows the full six-label classifier, but only `clear`, `near_mass_loss`, and `far_broadening` are populated in this scan window; the numbers in parentheses give the per-label counts for this panel.

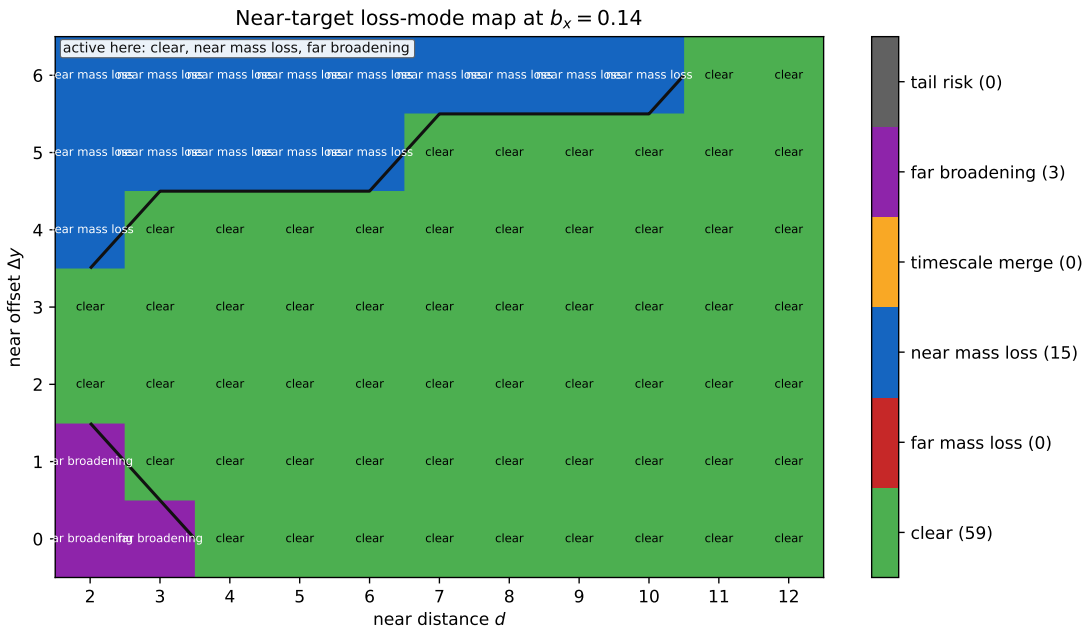


Figure 25: Near-target loss-mode map at $b_x = 0.14$. Increasing the forward bias shrinks the robust region mainly through the same near-branch failure mode; although the colorbar keeps all six labels for consistency, only three are activated in this panel as well.

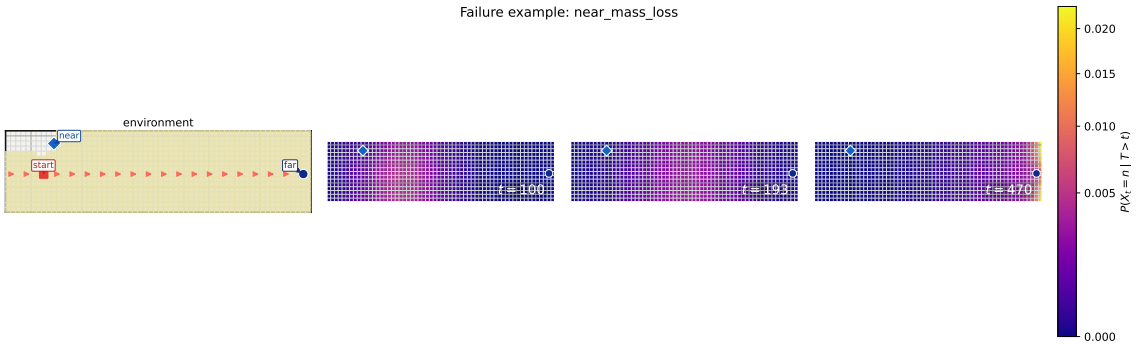


Figure 26: Failure exemplar at $(b_x, d, \Delta y) = (0.12, 2, 6)$, classified as `near_mass_loss`. The slow far branch remains, but the fast near branch loses enough weight that the clear-double case disappears.

The near-target position scan addresses the second question directly. Over the full $(d, \Delta y)$ map, 67/77 cases remain phase 2 at $b_x = 0.12$, while 59/77 remain phase 2 at $b_x = 0.14$. Across both slices the dominant failure mode is `near_mass_loss` (22 cases in total), with only six `far_broadening` cases and no `tail_truncation_risk` after the adaptive-horizon guard. The failure exemplar in Figure 26 makes the interpretation concrete: moving the near target too far off alignment destroys the fast branch before it destroys the slow branch.

| $b_x$ | clear $d$ interval | best $d$ | phase(best) | $P$(near) | $P$(far) |
|---|---|---|---|---|---|
| +0.00 | - | 3 | 0 | 0.892 | 0.108 |
| +0.02 | - | 2 | 0 | 0.603 | 0.397 |
| +0.04 | 3-9 | 3 | 2 | 0.444 | 0.556 |
| +0.05 | 2-10 | 2 | 2 | 0.393 | 0.607 |
| +0.06 | 2-10 | 2 | 2 | 0.355 | 0.645 |
| +0.07 | 2-10 | 2 | 2 | 0.326 | 0.674 |
| +0.08 | 2-10 | 2 | 2 | 0.301 | 0.699 |
| +0.09 | 2-10 | 2 | 2 | 0.281 | 0.719 |
| +0.10 | 2-10 | 2 | 2 | 0.264 | 0.736 |
| +0.11 | 2-10 | 2 | 2 | 0.248 | 0.752 |
| +0.12 | 2-10 | 2 | 2 | 0.235 | 0.765 |
| +0.14 | 2-10 | 2 | 2 | 0.212 | 0.788 |
| +0.16 | 2-10 | 3 | 2 | 0.206 | 0.794 |

Table 5: Background summary of the original $(d, b_x)$ scan: clear intervals and best near-target distance per $b_x$.

| spec | $b_x$ | $d$ | $\Delta y$ | phase | sep-score | valley/max | $p_{\min}$ | peak-ratio | peak-margin | min-margin | gate-pass | gate-slack | clear-score | showcase | $\varepsilon_{\text{split}}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| adaptive_refine | +0.12 | 4 | 1 | 2 | 2.38 | 0.007 | 0.350 | 0.170 | 0.070 | 0.070 | yes | 0.070 | 1.54 | 1.92 | $1.18 \times 10^{-9}$ |

Table 6: No-corridor clear-instance quality metrics (auto-selected from all candidate scans). We report `peak-margin`, `min-margin`, and strong-gate audit fields (`gate-pass`/`gate-slack`) to avoid near-threshold pseudo-clear picks, and include `showcase` for visual ranking consistency.

The clear-instance metrics (near/far masses, sep-score, valley/max, peak/min margins, showcase score, and splitting-committor closure error $\varepsilon_{\text{split}}$) are listed in Table 6. The selected showcase case comes from the adaptive-refinement round, $(b_x, d, \Delta y) = (0.12, 4, 1)$, with `peak-margin`=0.070, `min-margin`=0.070, and `showcase`=1.92. It passes the first robustness gate

$$\text{peak-margin} \geq 0.07, \qquad \text{min-margin} \geq 0.07, \qquad \text{valley}/\max \leq 0.10.$$

Its primary-gate audit reports `gate-pass=yes`, `gate-slack`=0.070 (minimum slack over the three thresholds), and 168 candidates in the first gate pool. We keep this point as the sharpest picture, but all mechanism claims in Track B are anchored at $(0.12, 2, 2)$.

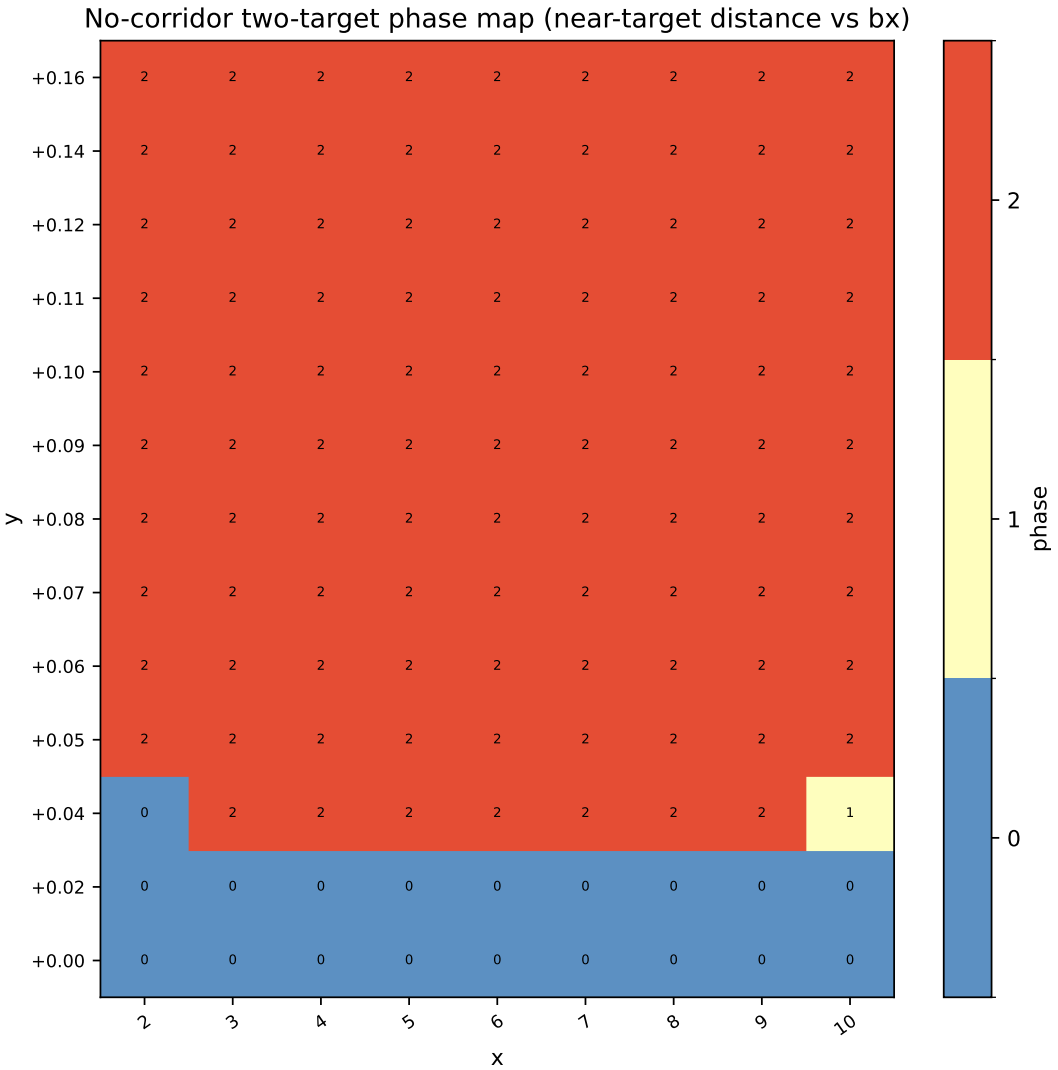


Figure 27: Coarse no-corridor phase map (near-target distance $d$ vs $b_x$) retained as a global reference. The colorbar is discrete: phase $\in \{0, 1, 2\}$ only.

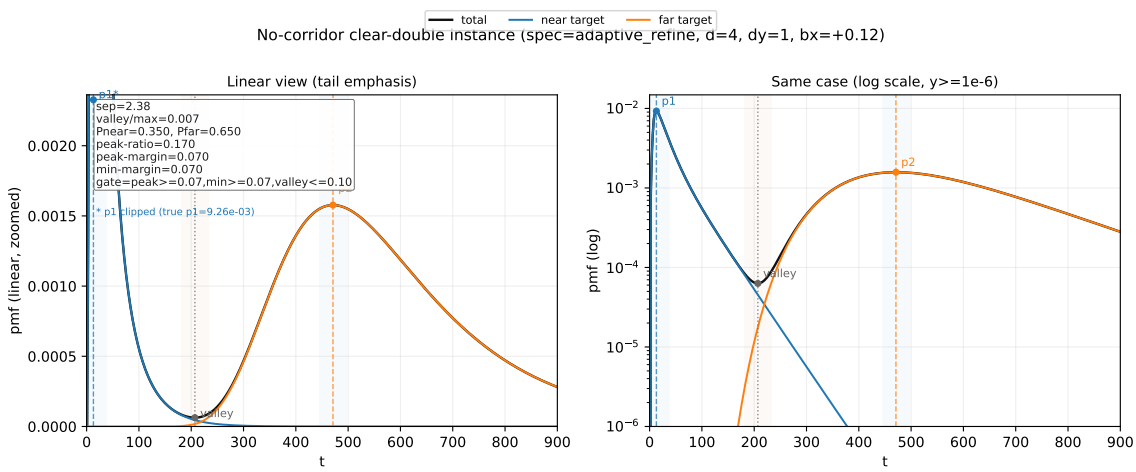


Figure 28: Dedicated no-corridor showcase clear-instance (maximum showcase score over all candidate scans): tail-emphasized linear panel plus log panel to keep valley/peak2 structure explicit.

# 4 Method Reference (Committor + 2x2 Classes)

### Reproducible Definition of Phase and sep-score

To make phase maps fully reproducible, we use the same criteria as in `src/vkcore/grid2d/rect_bimodality/cli.py`:

**(1) Two-peak detection (`find_two_peaks`)** Let $\tilde{f} = \text{smooth}_7(f)$ (window-7 smoothing). Candidate peaks satisfy

$$\tilde{f}(t) > \tilde{f}(t-1), \quad \tilde{f}(t) \geq \tilde{f}(t+1), \quad \tilde{f}(t) \geq 0.06 \max \tilde{f}.$$

Pair selection then enforces (default code thresholds)

$$t_{p2} - t_{p1} \geq 20, \ \text{peak_balance} \geq 0.08, \ \text{valley_drop} \geq 0.08, \ \text{minor_drop} \geq 0.005,$$

plus side-dominance tolerance (`dominance_tol`=0.02). Define

$$\text{has_double} := \mathbf{1}\{t_{p1}, t_{p2} \text{ both exist}\}.$$

These front-end conditions already form the first anti-fake filter:

- Smoothing $\tilde{f} = \text{smooth}_7(f)$ suppresses parity oscillations, staircase artifacts, and tiny numerical ripples, so discrete noise is not promoted into a second peak.
- The relative threshold $0.06 \max \tilde{f}$ removes very small tail bumps; if a would-be second peak is below 6% of the main peak, it is more naturally treated as tail roughness than as a genuine lobe.
- The time-gap condition $t_{p2} - t_{p1} \geq 20$ rejects shoulder structures on a single broad peak; without enough separation, two local maxima are usually just one lobe with a wrinkle.
- Candidate-level balance peak_balance $\geq 0.08$ rejects extremely weak secondary ripples, so a tiny bulge next to a main peak is not counted as a second peak.
- valley_drop $\geq 0.08$ enforces a nontrivial valley below the higher peak and removes shallow-notch pseudo-splitting.
- minor_drop $\geq 0.005$ further requires the valley to sit meaningfully below the weaker peak itself, specifically targeting near-flat plateaus or light waviness that can masquerade as a second peak.
- Side-dominance tolerance forces each selected peak to be near-dominant on its own side of the valley, which blocks the common shoulder-artifact failure mode.

So the later phase labels operate only after this first pass has already removed most obvious fake double peaks.

**(2) One-target phase definition** Define

$$\text{sep}_{\text{peaks}} := \frac{|t_{p2} - t_{p1}|}{w_{1/2}^{(1)} + w_{1/2}^{(2)}}, \quad \text{valley/max} := \frac{h_v}{\max(h_{p1}, h_{p2})}, \quad \text{peak_balance} := \frac{\min(h_{p1}, h_{p2})}{\max(h_{p1}, h_{p2})}.$$

Here $w_{1/2}^{(k)}$ is half-width at half-maximum of peak $k$ (computed on raw $f$), while $h_{p1}, h_{p2}, h_v$ are read from smoothed $\tilde{f}$. Then

$$\text{phase}_{\text{OT}} = \begin{cases} 2, & \text{has_double} = 1, \ \text{sep}_{\text{peaks}} \geq 1.0, \ \text{valley/max} \leq 0.35, \ \text{peak_balance} \geq 0.15, \\ 1, & \text{has_double} = 1 \text{ but the above clear-double test fails}, \\ 0, & \text{has_double} = 0. \end{cases}$$

The role of each threshold is:

- has_double = 1 is the basic prerequisite: if the front-end detector does not find two credible peaks, the case cannot be called clear-double.
- $\text{sep}_{\text{peaks}} \geq 1.0$ requires the distance between the two peak centers to be at least comparable to the sum of their half-widths; this blocks "one broad peak with a small wiggle" and heavily overlapping two-lobe shapes.
- valley/ max $\leq 0.35$ requires a sufficiently deep valley; this blocks shallow-notch or double-shoulder shapes that are visually close to a single peak.
- peak_balance $\geq 0.15$ requires the weaker peak to retain at least 15% of the stronger peak height; this blocks cases where the "second peak" is only a tiny bump trailing the main peak.

So for one-target runs, **phase=2** is a deliberately shape-based clear-double label: two peaks must be separated, the valley must be real, and the weaker peak must still be non-negligible.

**(3) Two-target phase definition** Define modal times and mode-separation score

$$t_{\text{mode}}^{\text{near}} := \arg\max_t f_{\text{near}}(t), \quad t_{\text{mode}}^{\text{far}} := \arg\max_t f_{\text{far}}(t),$$

$$\text{sep}_{\text{mode}} := \frac{|t_{\text{mode}}^{\text{far}} - t_{\text{mode}}^{\text{near}}|}{w_{1/2}^{\text{far}} + w_{1/2}^{\text{near}}}.$$

Define normalized splitting masses

$$P_{\text{near}} := \frac{\sum_t f_{\text{near}}(t)}{\sum_t f_{\text{any}}(t)}, \qquad P_{\text{far}} := \frac{\sum_t f_{\text{far}}(t)}{\sum_t f_{\text{any}}(t)}.$$

On the total law, define

$$\text{peak_ratio} := \frac{\min(h_{p1}, h_{p2})}{\max(h_{p1}, h_{p2})}, \qquad \text{valley/ max} := \frac{h_v}{\max(h_{p1}, h_{p2})}.$$

Then

$$\text{phase}_{\text{TT}} = \begin{cases} 2, & \text{has_double} = 1,\ \text{sep}_{\text{mode}} \geq 1.0,\ \min(P_{\text{near}}, P_{\text{far}}) \geq 0.15, \\ & \text{valley/ max} \leq 0.35,\ \text{peak_ratio} \geq 0.10,\ (\text{peak_ratio} - \text{valley/ max}) \geq 0.01, \\ 1, & \text{has_double} = 1 \text{ but the above clear-double test fails}, \\ 0, & \text{has_double} = 0. \end{cases}$$

The two-target gate is stricter than the one-target gate because here we can test not only total-curve shape but also the physical near/far branch split:

- has_double = 1 is still required at the level of the total FPT curve; if the total law does not even look two-peaked, the case cannot be clear-double.
- $\text{sep}_{\text{mode}} \geq 1.0$ compares the modal times and widths of $f_{\text{near}}$ and $f_{\text{far}}$ themselves, not just the total curve; this blocks cases where the total curve has a shallow notch even though the two physical branches do not have genuinely separated time scales.
- $\min(P_{\text{near}}, P_{\text{far}}) \geq 0.15$ forces both branches to carry nontrivial mass; this blocks "one real branch plus one tiny tail branch" from being promoted into a true double-peak claim.
- valley/ max $\leq 0.35$ again enforces a real valley in the total curve and rejects shallow-notch shapes.
- peak_ratio $\geq 0.10$ requires the weaker peak in the total law to stay above 10% of the stronger one; this blocks extremely small secondary bumps.
- (peak_ratio − valley/ max) $\geq 0.01$ further requires the weaker peak to stand measurably above the valley floor; this specifically blocks plateau-like or lightly wavy shapes where the "second peak" is barely above the intervening valley.

So mathematically, **phase=1** means "double peaks are detected, but at least one clear-double threshold is not met", while **phase=2** in the two-target setting should be read as a joint statement about total-curve shape *and* physically separated near/far branches.

## Adaptive Horizon Guard and Loss-Mode Rules

For Track B continuation scans we do not infer "the second peak disappeared" from a fixed horizon. We start from $t_{\max} = 12000$ and double it until both

$$\text{survival_tail} \le 10^{-12}, \qquad t_{\text{mode,far}} < 0.85\, t_{\max},$$

or until the hard cap $t_{\max} = 48000$ is reached. Any row that still violates the guard is labeled `tail_truncation_risk`. The ordered loss-mode classifier is then applied as

`clear` → `far_mass_loss` → `near_mass_loss` → `timescale_merge` → `far_broadening` → `tail_truncati`

with the following concrete meanings:

- `clear`: the row still satisfies the full two-target clear-double gate, i.e., phase=2.
- `far_mass_loss`: $P_{\text{far}} < 0.15$, so the slow far branch loses too much total mass to support the second peak.
- `near_mass_loss`: $P_{\text{near}} < 0.15$, so the fast near branch loses mass first and the early peak collapses.
- `timescale_merge`: both masses remain nontrivial, but $\text{sep}_{\text{mode}} < 1.0$ or $t^{\text{far}}_{\text{mode}} - t^{\text{near}}_{\text{mode}} < 60$, meaning the near/far conditional timescales have merged and the valley is no longer sharp.
- `far_broadening`: none of the earlier rules fires, but the far branch has broadened enough to smear out the slow peak; physically, the late branch still exists in mass but no longer produces a clean peak top.
- `tail_truncation_risk`: even at $t_{\max} = 48000$ the horizon guard is not passed, so a missing second peak must still be treated as a time-window truncation risk rather than a physical loss.

So a lost case is first interpreted in terms of branch mass and branch width before being called a classification failure. In the current start and near-target continuation scans, all rows pass the horizon guard; the dominant loss mode is therefore physical (`near_mass_loss`), not numerical truncation. In the present $(d, \Delta y)$ window, only `clear`, `near_mass_loss`, and `far_broadening` are actually populated, so the maps display only three colors even though the unified classifier keeps six labels.

## Committor Mechanism (Detailed)

Let the hitting time be $\tau_C := \inf\{t \ge 0 : X_t \in C\}$. On the discrete Markov chain with transition matrix $P$, let $A$ be the start basin (here a strip near the start) and $B$ be the absorbing target set. Define

$$q_i := \mathbb{P}_i(\tau_B < \tau_A),$$

which solves the discrete Dirichlet boundary-value problem

$$q_i = 0 \ (i \in A), \qquad q_i = 1 \ (i \in B), \qquad q_i = \sum_j P_{ij} q_j \ (i \notin A \cup B).$$

With $U = \Omega \setminus (A \cup B)$, this is equivalent to the linear system

$$(I - P_{UU})\, q_U = P_{UB}\, \mathbf{1}.$$

For this absorbing setup the solution is unique, so $q$ is a dynamics-determined progress coordinate (not an ad hoc geometric proxy). We use $q$ as the progress coordinate and set $q^* = 0.5$ to define the commitment surface

$$\Sigma_{q^*} := \{i : \ q_i \ge q^*\}.$$

Moreover, when $\rho(P_{UU}) < 1$, the Neumann expansion gives

$$q_U = (I - P_{UU})^{-1} P_{UB}\mathbf{1} = \sum_{n\geq 0} P_{UU}^n P_{UB}\mathbf{1}.$$

So $q_U$ is the cumulative contribution of paths that spend $n$ interior steps before absorption into $B$, which provides a direct linear-system bridge to the renewal failure-retry interpretation used later: early peak mass is small-$n$ dominated, while the slow peak accumulates from larger-$n$ loops.

### L×R Events and Recross Counts

For a trajectory $X_t$ up to $\tau_B$, define membrane-cross count

$$N_{\text{mem}} := \sum_{t=0}^{\tau_B - 1} \mathbf{1}\{(X_t, X_{t+1}) \in \mathcal{E}_{\text{mem}}\},$$

and commitment-to-start recross count

$$N_{\text{ret}} := \sum_{t=0}^{\tau_B - 1} \mathbf{1}\{X_t \in \Sigma_{q^*},\ X_{t+1} \in A\}.$$

Then use binary labels

$$L = \mathbf{1}\{N_{\text{mem}} \geq 1\}, \qquad R = \mathbf{1}\{N_{\text{ret}} \geq 1\},$$

which yields four classes

$$(L, R) \in \{(0,0), (0,1), (1,0), (1,1)\},$$

implemented as `L0R0/L0R1/L1R0/L1R1` (Figure 14).

### Peak/Valley Mechanism as Renewal Decomposition

Let $f(t)$ be the FPT law. With direct-success kernel $g(t)$ and fail-return kernel $h(t)$,

$$f(t) = g(t) + (h * f)(t) = \sum_{k\geq 0} (h^{*k} * g)(t).$$

Here $k$ is the number of failed attempts before success. Bimodality appears when the $k = 0$ (near one-shot) and $k \geq 1$ parts separate in both timescale and mass. Window fractions

$$w_{\ell r}(W) = \frac{\sum_{t\in W} f_{\ell r}(t)}{\sum_{t\in W} f(t)}$$

make the transfer explicit: peak1 is mainly `L0R0`; valley remains `L0R0`-major with a clear rise of `L1R0`; and peak2 is `L0R1`-dominant, with `L1R1` secondary (Figure 13, Table 3).

### Closed-Form Bridge from Linear System to FPT Kernels

For the no-corridor two-target branch, let $U$ denote interior transient states, $e_s$ the interior basis vector at the start, and

$$Q := P_{UU}, \qquad R_{\text{near}} := P_{U,\partial_{\text{near}}}, \qquad R_{\text{far}} := P_{U,\partial_{\text{far}}}.$$

Then the first-hit pmfs are

$$f_{\text{near}}(t) = e_s^\top Q^{t-1} R_{\text{near}}\mathbf{1}, \qquad f_{\text{far}}(t) = e_s^\top Q^{t-1} R_{\text{far}}\mathbf{1}, \qquad t \geq 1.$$

Hence the far-hit mass is

$$P_{\text{far}} = \sum_{t\geq 1} f_{\text{far}}(t) = e_s^\top (I-Q)^{-1} R_{\text{far}} \mathbf{1}.$$

This gives the exact bridge between time-domain decomposition and the linear-system view used by committors.

### Conditioned Kernel (Doob Transform) and Peak Transfer

On interior states with $q_{\text{far}}(i) > 0$, define

$$\widetilde{P}_{ij}^{\text{far}} := \frac{P_{ij}\, q_{\text{far}}(j)}{q_{\text{far}}(i)}.$$

This is the Doob $h$-transform of $P$ by $h = q_{\text{far}}$, i.e., the exact dynamics conditioned on eventually hitting *far* before *near*. In this view, the near branch is the complementary conditioned dynamics through $q_{\text{near}} = 1 - q_{\text{far}}$. Therefore, the window transfer from peak1 to peak2 can be interpreted as a measure transfer between two conditioned ensembles: early-time mass is dominated by short successful excursions, while late-time mass is amplified by loops that repeatedly leave and re-enter low-$q_{\text{far}}$ regions before final absorption. This provides a direct dynamical meaning for the observed renewal split.

### Spectral View: Origin of Fast/Slow Timescales

On the interior kernel $Q$, write the (generalized) spectral expansion

$$Q^{t-1} = \sum_k \lambda_k^{t-1} r_k \ell_k^\top, \qquad |\lambda_1| > |\lambda_2| \geq \cdots .$$

Then

$$f_{\text{near}}(t) = \sum_k c_k^{\text{near}} \lambda_k^{t-1}, \qquad f_{\text{far}}(t) = \sum_k c_k^{\text{far}} \lambda_k^{t-1}.$$

Modes with smaller $|\lambda_k|$ decay quickly and dominate early windows, while the leading mode $\lambda_1$ controls the slow tail and late peak. If near/far branches project differently on these modes, we naturally obtain "near-dominant early window" versus "far-dominant late window" transfer. This is fully consistent with the renewal picture: peak1 is short-stay/low-loop mass, while peak2 is quasi-stationary release from the leading mode. Equivalently, the conditioned hazard

$$h_{\text{far}}(t) := \frac{f_{\text{far}}(t)}{S(t-1)}$$

increases over time when the far channel progressively takes over late windows.

### Splitting Committor Closure for Two Targets

On the same system, define

$$q_{\text{far}}(i) := \mathbb{P}_i(\tau_{\text{far}} < \tau_{\text{near}}), \qquad q_{\text{near}}(i) = 1 - q_{\text{far}}(i).$$

Its boundary-value problem is

$$q_{\text{far}} = 0 \text{ on near}, \qquad q_{\text{far}} = 1 \text{ on far}, \qquad q_{\text{far}}(i) = \sum_j P_{ij}\, q_{\text{far}}(j) \text{ on interior}.$$

This gives a direct closure with the exact decomposition $f = f_{\text{near}} + f_{\text{far}}$: $q_{\text{far}}(x_s)$ should match $P_{\text{far}}$. Define

$$\varepsilon_{\text{split}} := \left| q_{\text{far}}(x_s) - \sum_{t \geq 1} f_{\text{far}}(t) \right| = \left| q_{\text{far}}(x_s) - P_{\text{far}} \right| .$$

Moreover, let $\tau := \tau_{\text{near}} \wedge \tau_{\text{far}}$ and define

$$M_t := q_{\text{far}}(X_{t \wedge \tau}).$$

Because $q_{\text{far}}$ is harmonic on interior states and fixed to 0/1 on near/far boundaries, $M_t$ is a bounded martingale. Optional stopping then gives

$$q_{\text{far}}(x_s) = \mathbb{E}[M_\tau] = \mathbb{P}_{x_s}(\tau_{\text{far}} < \tau_{\text{near}}) = P_{\text{far}}.$$

So $\varepsilon_{\text{split}}$ is theoretically zero; numerically, its scale should track the Bellman-residual scale, making it a joint diagnostic for mechanism closure and solver stability.

| case | $q_{\text{far}}(x_s)$ | $P_{\text{far}}$ | $\varepsilon_{\text{split}}$ | $\lVert r_{\text{int}} \rVert_\infty$ | $\lVert r_{\text{near}} \rVert_\infty$ | $\lVert r_{\text{far}} \rVert_\infty$ |
|---|---|---|---|---|---|---|
| representative | 0.676009 | 0.676009 | $1.14 \times 10^{-9}$ | $9.93 \times 10^{-12}$ | 0 | 0 |
| clear-instance | 0.650264 | 0.650264 | $1.18 \times 10^{-9}$ | $9.93 \times 10^{-12}$ | 0 | 0 |

Table 7: Splitting-committor closure QC for no-corridor two-target runs (physics anchor, listed as "representative" in the legacy table, plus dedicated clear-instance): we report $q_{\text{far}}(x_s)$, $P_{\text{far}}$, $\varepsilon_{\text{split}}$, and interior/boundary residual norms.

Table 7 shows that both routes satisfy $q_{\text{far}}(x_s) \approx P_{\text{far}}$, with $\varepsilon_{\text{split}}$ already at the $10^{-9}$ scale for both rows, while interior Bellman residuals stay at the $10^{-11}$ scale. This closes the mechanism loop behind the fast/slow peak split for both the physics-anchor route and the showcase-quality clear-instance route. We track these residuals together with $\varepsilon_{\text{split}}$ as automatic QC scalars in continuous optimization runs.

### Committor Implementation in This Report

1. Build the exact transition kernel for each anchor case (including laziness, drift, membrane permeability, and reflecting boundaries).
2. Use $A = \{x \leq x_s + 1,\ |y - y_s| \leq 1\}$ and $B = \{\text{target}\}$, and solve $q$ via fixed-point iteration with tolerance $10^{-11}$; record interior Bellman residual and boundary mismatch as quality diagnostics, and render the basin/commitment schematic in Figure 6.
3. Use exact augmented-state recursion on $(x, \ell, s, r)$ (not MC): $\ell$ tracks leak, $s$ tracks whether $\Sigma_{q^*}$ has been visited, and $r$ tracks strict recross (return to $A$ after visiting $\Sigma_{q^*}$).
4. Report $q^* = 0.5$ as default and additionally provide $q^* \in \{0.4, 0.5, 0.6\}$ robustness results (Table 4).
5. For no-corridor two-target cases, keep the strict identity $f = f_{\text{near}} + f_{\text{far}}$ as level-1 decomposition, export the full $q_{\text{far}}$ field plus the $q_{\text{far}} = 0.5$ mask, and then interpret windows with committor-consistent labels.

## 5 Takeaways and Next Steps (Committor-Centric)

### Implementation Summary

1. Membrane corridor track: clear-double persists in both symmetric and asymmetric representatives; under strict recross, peak1 is mostly `L0R0`, valley gains `L1R0`, and peak2 is mostly `L0R1`.

2. No-corridor two-target track: Track B is now explicitly split into a physics anchor $(0.12, 2, 2)$ and a showcase clear-instance $(0.12, 4, 1)$. The physics anchor gives $P_{\text{near}}/P_{\text{far}} \approx 0.324/0.676$ with $(t_{p1}, t_v, t_{p2}) = (7, 196, 469)$, while the showcase case keeps the strongest visual margins and passes the first robustness gate (peak-margin $\geq 0.07$, min-margin $\geq 0.07$, valley/ max $\leq 0.10$). We additionally export strong-gate audit scalars (`gate-pass`/`gate-slack`), first-gate candidate count (168), the start-scan continuation CSV, and the $(d, \Delta y)$ loss-mode CSV.
3. Continuation results now answer the two physics questions directly: in the `source_only_x` start scan only 4/11 positions remain `clear`, whereas the comoving control `source_plus_near_x` stays `clear` for all 11 positions; in the $(d, \Delta y)$ near-target scan the dominant failure mode is `near_mass_loss` (22 rows total), with no surviving truncation-risk rows after the adaptive horizon guard.
4. Methodologically, committor is now fully operational in the pipeline; splitting committor is numerically closed against exact far-hit mass for both the physics anchor and the clear-instance, and we jointly monitor $\varepsilon_{\text{split}}$ plus committor residual diagnostics in automatic QC (current $\varepsilon_{\text{split}}$ scale $\sim 10^{-9}$ for both rows).
5. Operationally, the current full local regression (codegen + CN/EN build checks) takes about 70 s per round (around 1.2 minutes), so it stays in the local-first regime ($<$ 10 minutes); we switch to Isambard only when projected single-round runtime exceeds 20 minutes.

### Next Directions (All Grounded in Committor)

1. Extend the current physics-anchor continuation from 1D start scans and two $b_x$ slices to the full $(d, \Delta y, b_x)$ continuation surface, while keeping the same loss-mode bookkeeping.
2. Run $q^*$ sensitivity analysis (e.g., 0.4/0.5/0.6) and report peak/valley class stability.
3. Turn the exported splitting-committor fields and window summaries into threshold-margin maps and trajectory-family atlases for the whole two-target scan:

$$q_{\text{far}}(x) = \mathbb{P}_x(\tau_{\text{far}} < \tau_{\text{near}}),$$

   while continuing to use both $\varepsilon_{\text{split}} = |q_{\text{far}}(x_s) - P_{\text{far}}|$ and `min-margin`/`gate-slack` as automatic quality checks.
4. Compare top/bottom membrane asymmetry versus directional asymmetry (inside/outside permeability mismatch) in a unified committor geometry view.
5. Keep continuous build checks active (codegen + EN/CN LaTeX) and enforce routing policy in keepalive: run local for short rounds ($<$ 10 minutes), and use isbard doctor/auth + submit/status/fetch for large scans (projected round $>$ 20 minutes), with automatic local-smoke fallback on remote failures.
6. For every remote round, always record `JOB_ID`, `REMOTE_DIR`, key artifact paths, and explicit next actions in the run summary.

## Reproducibility

```
cd research/reports/grid2d_membrane_near_target
../../../.venv/bin/python code/membrane_near_target_report.py
cd manuscript
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -auxdir=build -emulate-aux-dir grid2d_membrane_near_target_en.tex
```